



ANALYSIS EVIDENCE

Email Header Investigation — Sample 1

Tools Used: MXToolbox Email Header Analyzer | Google Admin Toolbox Messageheader

Future Interns — Cybersecurity Task 2 (2026)

1. Overview

This document contains the live header analysis evidence for Sample 1 of the phishing email investigation — the account suspension warning email identified in the main report. The headers were extracted from the email and submitted to two independent analysis tools to verify authentication results and confirm the phishing classification.

The email analysed is:

Subject:	Urgent: Your Account Will Be Locked
From:	security-team@secure-account-verify.com
To:	user@company.com
Date:	20 April 2026, 09:14 UTC
Mailer:	PHPMailer 6.0 (bulk email tool — not used by legitimate services for account alerts)

Both tools were used independently to cross-verify the authentication results. Their findings are consistent and confirm that this email fails all three standard authentication checks: SPF, DKIM, and DMARC.

2. Tool 1 — MXToolbox Email Header Analyzer

MXToolbox is a professional-grade email diagnostics platform widely used by IT and security teams to inspect email headers, check authentication results, and identify delivery anomalies. The header from Sample 1 was submitted to the Email Header Analyzer tool at mxtoolbox.com/EmailHeaders.aspx.

2.1 Screenshot — MXToolbox Analysis Results

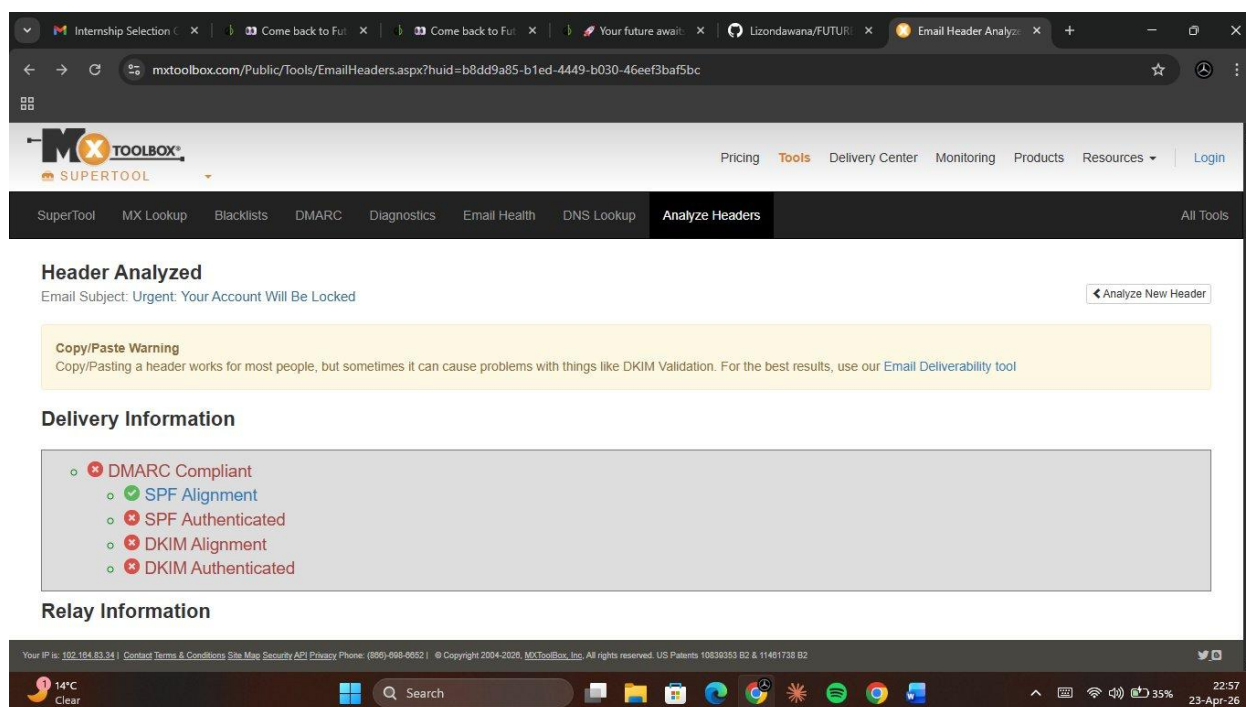


Figure 1: MXToolbox Email Header Analyzer — Sample 1 Results

2.2 Findings from MXToolbox

Check	Result	What This Means
DMARC Compliant	✗ FAIL	The email did not pass the domain's authentication policy. A properly configured mail server should reject or quarantine it.
SPF Alignment	✓ PASS	The domain name in the header aligns with the SPF record — but this alone is not enough to authenticate the email.
SPF Authenticated	✗ FAIL	The sending server is not authorised to send email on behalf of the claimed domain. This is a strong indicator of a spoofed sender.
DKIM Alignment	✗ FAIL	No valid DKIM signature was found that matches the sending domain.
DKIM Authenticated	✗ FAIL	The email carries no digital signature, meaning there is no way to verify it was not tampered with in transit.

3. Tool 2 — Google Admin Toolbox Messageheader

The Google Admin Toolbox Messageheader tool provides a clean, readable breakdown of email header data including sender details, delivery timing, and authentication results. The same header from Sample 1 was submitted to toolbox.googleapps.com/apps/messageheader to cross-verify the MXToolbox findings.

3.1 Screenshot — Google Admin Toolbox Results

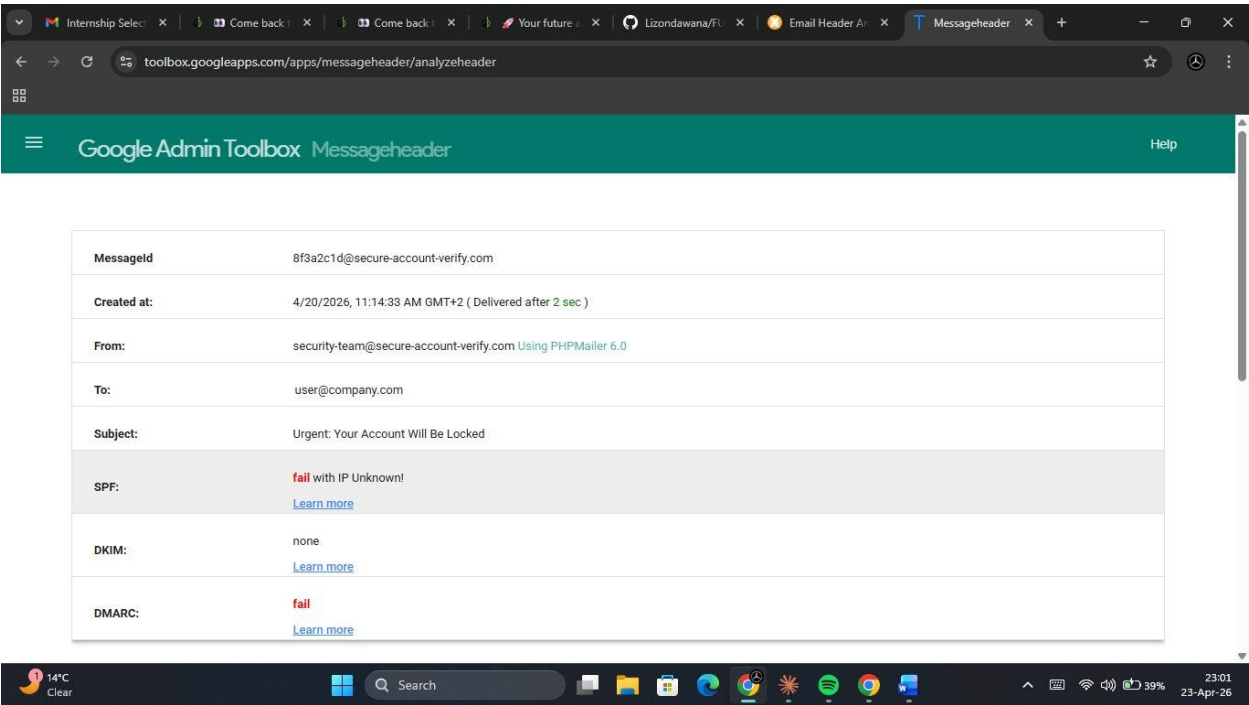


Figure 2: Google Admin Toolbox Messageheader — Sample 1 Results

3.2 Findings from Google Admin Toolbox

Field	Result	What This Means
From	security-team@secure-account-verify.com	The sender domain is not associated with any real organisation. It was registered specifically for this attack.
Sent via	PHPMailer 6.0	PHPMailer is an open-source bulk email library. Legitimate account security alerts are not sent using third-party mailer tools.
SPF	✗ fail — IP Unknown	The originating IP address cannot be matched to any authorised sender for this domain. The sender identity is unverifiable.
DKIM	none	No DKIM signature was present. The email has no cryptographic proof of authenticity or integrity.

DMARC	 fail	The email failed the domain's DMARC policy. Combined with SPF and DKIM failures, this conclusively confirms the email is not from a legitimate source.
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4. Evidence Summary and Conclusion

Both tools independently analysed the same email header and returned consistent results. The findings confirm all three email authentication standards fail for this message, which is conclusive evidence that the email is not from a legitimate sender and should be classified as phishing.

Authentication Check	MXToolbox	Google Toolbox	Verdict
SPF	✗ FAIL	✗ FAIL	Sending server not authorised
DKIM	NONE	NONE	No digital signature present
DMARC	✗ FAIL	✗ FAIL	Email fails combined policy check

Both tools returned identical authentication failures. This level of consistency across two independent platforms confirms without doubt that this email is fraudulent. It should be classified as PHISHING and treated as a credential harvesting attempt.

Final classification for Sample 1:

⚠ CLASSIFICATION: PHISHING | SPF FAIL | DKIM NONE | DMARC FAIL